

JavaScript Fundamentals Assignment

Section 1: Data Types in JavaScript

Question 1.1: What will be the output of the following code?

javascript

```
console.log(typeof 42);  
console.log(typeof "hello");  
console.log(typeof true);  
console.log(typeof undefined);  
console.log(typeof null);
```

Question 1.2: Identify the data type of each variable:

javascript

```
let a = 25;  
let b = "JavaScript";  
let c = false;  
let d;  
let e = 3.14;
```

Question 1.3: What will be the output?

javascript

```
console.log(typeof (5 + "3"));  
console.log(typeof (5 - "3"));  
console.log(typeof ("5" * 2));
```

Section 2: var vs let vs const

Question 2.1: What will be the output of this code?

javascript

```
var x = 10;  
var x = 20;  
console.log(x);
```

Question 2.2: What will happen when you run this code?

javascript

```
let y = 30;  
let y = 40;  
console.log(y);
```

Question 2.3: What will be the output?

javascript

```
const z = 50;  
z = 60;  
console.log(z);
```

Question 2.4: What will be the output of this code block?

javascript

```
{  
  var a = 100;  
  let b = 200;  
  const c = 300;  
}  
console.log(a);  
console.log(b);  
console.log(c);
```

Question 2.5: What will be the output?

javascript

```
console.log(p);  
var p = 15;  
console.log(q);  
let q = 25;
```

Section 3: ASCII Values

Question 3.1: What will be the output of the following code?

javascript

```
console.log("A".charCodeAt(0));  
console.log("z".charCodeAt(0));  
console.log("0".charCodeAt(0));
```

Question 3.2: What will be the output?

javascript

```
console.log(String.fromCharCode(65));  
console.log(String.fromCharCode(97));  
console.log(String.fromCharCode(48));
```

Question 3.3: What will be the result?

javascript

```
let char1 = "B";  
let char2 = "A";  
console.log(char1.charCodeAt(0) - char2.charCodeAt(0));
```

Section 4: String Concatenation and Number Operations

Question 4.1: What will be the output?

javascript

```
console.log("Hello" + " " + "World");  
console.log("5" + 3);  
console.log(5 + "3");  
console.log("5" + "3");
```

Question 4.2: What will be the result of these operations?

javascript

```
console.log(10 + 20);  
console.log(15 - 7);  
console.log(6 * 8);  
console.log(2 ** 3);
```

Question 4.3: What will be the output?

javascript

```
console.log("10" - 5);  
console.log("15" * 2);  
console.log("8" / 4);  
console.log("2" ** 3);
```

Question 4.4: What will be the result?

javascript

```
let num1 = 12;  
let num2 = 8;  
console.log(num1 + num2);  
console.log(num1 - num2);  
console.log(num1 * num2);  
console.log(num1 ** 2);
```

Section 5: Increment and Decrement Operators

Question 5.1: What will be the output?

javascript

```
let a = 5;  
console.log(a++);  
console.log(a);
```

Question 5.2: What will be the output?

javascript

```
let b = 8;  
console.log(++b);  
console.log(b);
```

Question 5.3: What will be the output?

javascript

```
let c = 10;  
console.log(c--);  
console.log(c);
```

Question 5.4: What will be the output?

javascript

```
let d = 15;  
console.log(--d);  
console.log(d);
```

Question 5.5: What will be the final value of x?

javascript

```
let x = 20;
```

```
x++;
```

```
++x;
```

```
x--;
```

```
console.log(x);
```